Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/04/2022
PI Name:	Yoo, Seung-Schik Ph.D.		
Project Title:	Wearable Modular Focused Ultrasound Systems for Non-Invasive Stimulation of the Human Brain During Deep Space Exploration		
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PI Email:	yoo@bwh.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-9464
Organization Name:	Brigham and Women's Hospital		
PI Address 1:	Neuromodulation and Tissue Engineering Laboratory		
PI Address 2:	221 Longwood Ave, EBRC Rm 521b		
PI Web Page:			
City:	Boson	**State:**	MA
Zip Code:	02115	**Congressional District:**	7
Comments:			
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Wonhye Ph.D. (Brigham and Women's Hospital) Weisholtz, Daniel M.D. (Harvard Medical School)		
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Task Description:	Deep space exploration will cast unprecedented and multi-faceted health challenges for crew members. Limited access to resources calls for non-pharmacological alternatives to improve the performance and wellness of astronauts in the midst of stressful environments during a long space flight mission. Non-invasive regional brain stimulation techniques may expand options that can benefit their mental as well as physical health. Focused ultrasound (FUS) techniques enable the delivery of highly-focused (with a focal size measuring a few millimeters) acoustic energy to biological tissue in a non-invasive fashion. A series of recent studies, including our own, have found that transcranial FUS (tFUS) sonication to the brain, given in batches of pulses at a low intensity (below the threshold for heat generation or mechanical damage to the tissue), can reversibly modulate the excitability of regional brain tissue without elevating its temperature. We have also demonstrated that FUS can modulate the degree of neural conduction through myelinated nerves and can elicit tactile sensations by directly stimulating the nerve endings, suggesting its further applicability to the peripheral nervous system. Based on the advantages over existing brain stimulation modalities, especially in terms of the spatial selectivity and depth penetration, tFUS technique is anticipated to provide greater flexibility in non-invasive neuromodulation. In addition to the implementation and optimization of the technique for a past decade, we have developed light-weight wearable FUS transducers and their control systems, augmented with image-guidance for targeting the sonication focus to individual neuroanatomy. On-site computer simulation is warranted to characterize the acoustic propagation through the skull, informing the user of the focal location and its intensity. The proposed ground-based study is intended to advance our knowledge on neuromodulatory potentials of tFUS while addressing crucial technical innovations necessary to achieve the accurate delivery of the acoustic focus to specific brain areas (both cortical and subcortical), with specifications suitable for its routine use during deep space exploration. We will develop a modular tFUS system equipped with optical-based individual image-guidance and on-site numerical simulation of the acoustic propagation. The FUS transducer will be made wearable, adopting a light-weight ergonomic headgear configuration. Prior to its use in modulating brain neural circuitries associated with performance-enhancing cognitive functions, we intend to examine the operating characteristics and safety of the technique in stimulating the primary somatosensory (S1) area and its thalamic projection (i.e., ventral posterolateral nucleus-VPL) among healthy humans. The range of sonication parameters that stimulate these brain regions at the lowest possible acoustic intensity will be examined via closed-loop monitoring of the electrophysiological responses (electroencephalograph sensory evoked potentials - SEP) and through subject self-reporting. Computer-generated, randomized/balanced inclusion of passive and active sham FUS conditions will provide a double-blind experimental design. We will assess the safety of the procedure through neuroradiological and neurological evaluations performed at variable time points, covering acute, delayed, and long-term periods after the sonication.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study was intended to advance our knowledge regarding neuromodulatory potentials of tFUS while addressing crucial technical innovations for deep space exploration. We developed a tFUS system using single-element transducer configuration (rather than a more error-prone multi-array transducer) for the reliable operation in spaceflight. A reconfigurable ergonomic headgear allowed for sonicating both cortical and deep brain structures. Light-weight, wearable FUS-enabled headgear and its adaptation to optical-based image-guidance were particularly suitable as a translational platform for human applications. Development of embedded real-time numerical acoustic simulation software also served as a critical innovation to ensure the accuracy of sonication in a spaceflight environment. All hardware components were prepared in modular configurations for maintenance, replacement, and troubleshooting in space. These state-of-the art technical innovations were used to subsequently demonstrate the efficacy and safety of the tFUS-mediated, non-invasive brain stimulation. We demonstrated the utility of our tFUS system in stimulating the primary sensory (corresponding to the unilateral hand area representation) and its thalamic projections among healthy humans as well as for stimulating primary motor (corresponding to the unilateral hindlimb) and thalamus in awake animals. Because there is very limited data regarding tFUS neuromodulation in humans to date, a range of pulse duration sonication parameters that transiently modulate the function of the targeted neural circuitries were investigated. We found (1) the existence of specific pulse duration that yielded effective stimulation (which was found to be similar across species), (2) successfully stimulated the targeted brain areas, and (3) demonstrated the safety of the technique through comprehensive histological analysis of the ovine brain and extensive neurological assessment of the human volunteers. From human study using functional connectivity analysis, we also found, for the first time, that (4) the sonication may yield long-lasting neuromodulatory effects beyond the sonication itself, which may generate therapeutic effects via neural plasticity. Therefore, our study not only conferred firm technical foundations for stimulating somatosensory circuits among healthy human volunteers, but also established initial safety profiles prior to its ultimate use in spaceflight.
	Deep space exploration will cast unprecedented and multi-faceted health challenges for crew members. Limited access to resources during long spaceflight missions calls for non-pharmacological alternatives for improving the performance and wellness of astronauts. Non-invasive regional brain stimulation techniques may expand options that can benefit both the mental and physical health of astronauts. Focused ultrasound (FUS) techniques enable the delivery of highly-focused (with a focal size measuring a few millimeters) acoustic energy to biological tissue in a non-invasive fashion. Transcranial FUS (tFUS) sonication to the brain, given in batches of pulses at a low intensity (below the threshold for heat generation or mechanical damage to the tissue), can reversibly modulate the excitability of regional brain tissue without elevating its temperature. Based on the advantages over existing brain stimulation modalities, especially in terms of the spatial selectivity and depth of penetration, the tFUS technique is anticipated to provide greater flexibility in non-invasive neuromodulation than existing brain stimulation modalities. During the Year 1 of the project, we completed the development of a modular, low-power tFUS system, equipped with image-guidance capability and an on-site numerical simulation of the acoustic propagation, aimed for use during deep space exploration. The on-site computer simulation is warranted to characterize the acoustic propagation through the skull, informing the user of the focal location and intensity. We characterized the device in terms of its electrical and acoustical specifications and submitted the Final Design Specification to the Translational Research Institute for Space Health (TRISH) 6 months ahead of schedule. The FUS transducer was made wearable, adopting a light-weight ergonomic headgear configuration. The performance of the tFUS system and the integrated numerical simulation were assessed in terms of (1) spatial accuracy, (2) in situ acoustic pressure, and (3) shape of focus by using human skull phantoms in degassed water, and all surpassed the design goals. All staff were trained on the device operation following

Task Progress:	the standard operating procedure (SOP) established. During the Year 2 of the project, we received Food and Drug Administration (FDA) Study Risk Determination as a non-significant risk device on Feb 25, 2020. During the pandemic, which inevitably slowed down the experiments in humans, we obtained the approval from the TRISH to change the study objectives to test the device among large animals (n=8) and examined the efficacy and safety of the technique in stimulating the primary motor (M1) area and its thalamic projection. The range of sonication parameters, especially the pulse duration (PD), that stimulate these brain regions was examined via closed-loop monitoring of the electrophysiological responses (electromyograph/EMG). Upon receiving Institutional Review Board (IRB) approval on August 19, 2020, we decided to further examine the effects of the same experimental variables in stimulating the primary somatosensory (S1) area and its thalamic projection (i.e., ventral posterolateral nucleus-VPL) among healthy humans (n=8) via monitoring of the electroencephalograph evoked potentials (EP) as well as subject self-reporting. Computer-generated, randomized/balanced passive and active sham FUS conditions provided a double-blind experimental design. From the ovine model study, the group-averaged electromyography (EMG) responses from both hindlimbs across the experimental conditions revealed selective responses from the hindlimb contralateral to sonication while the use of 0.5 and 1 ms PD (given at 1,400 and 700 Hz PRF, respectively) generated higher EMG signal amplitude and response rates compared the use of 2 ms PD. Post-sonication behavioral observation and histological assessment performed within 24 h and after 1 month after the sonication did not reveal any abnormalities. From humans, we found bilateral EPs were observed from unilateral stimulation of the brain regions (corresponding to the somatosensory areas of the non-dominant hand), while showing higher EP amplitude changes elicited from the use of 0.5 ms PD. The results are in good agreement with the ovine study, which suggests that presence of superior simulation efficiency across species. Based on the functional connectivity (FC) analysis of resting state - functional magnetic resonance imaging (rs-fMRI) signals, we also identified that tFUS simulation enhanced the FC among motor-related circuits at least one hour, which suggests its potential utility in inducing neural plasticity. The changes in FC completely returned to baseline state one month after the stimulation. Through neuroradiological and neurological evaluations performed at variable time points, we also found the technique is safe. In summary, our ground-based study advanced our knowledge of the neuromodulatory potential of tFUS while addressing crucial technical innovations necessary to achieve the accurate placement of the acoustic focus in specific brain areas (both cortical and subcortical) with specifications suitable for routine use during deep space exploration. To our knowledge, this was the first study to demonstrate the differential excitatory efficacy of FUS neuromodulation based on different sonication pulsing schemes in large animals and humans, without confounding factors from anesthesia. As neuromodulatory FUS also has the capability to suppress regional brain activity, further insight on the effects of pulsing parameters on the subjects is desired. Neuromodulatory capability of FUS would incite a new mode of neurotherapeutics for various neurological and psychiatric disorders, and further probing of the effects of pulsing parameters in humans is desirable.
Bibliography Type:	Description: (Last Updated: 08/17/2023)
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